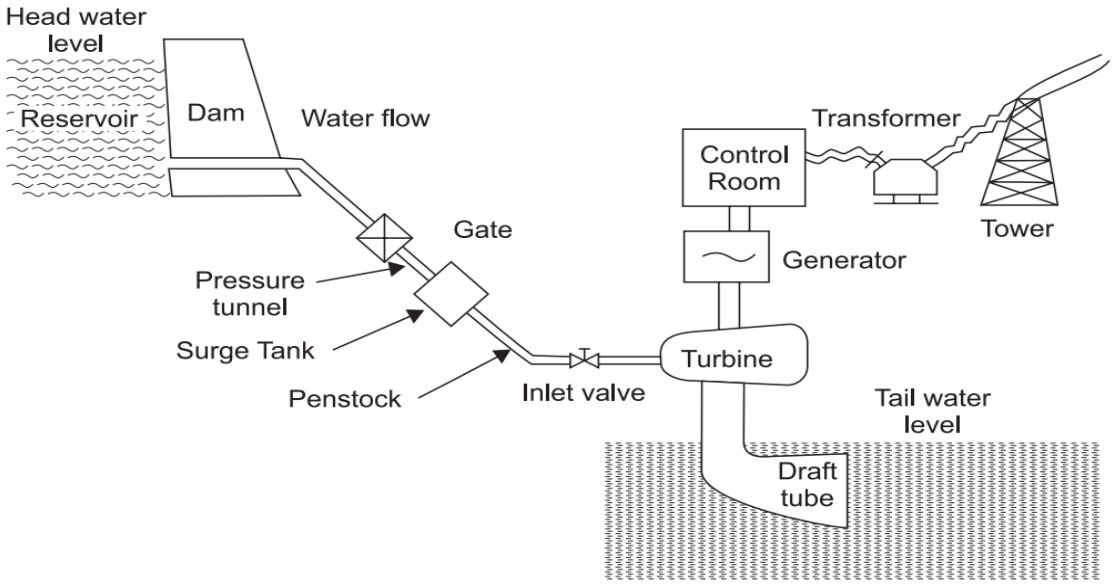
UNIT 3

Energy Resources, Electricity Bill and Safety Measures:
Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Thermal, Solar and Wind power generation. Calculation of electricity bill for domestic appliances. Working principle of Fuse and Miniature circuit breaker (MCB). Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

1. Explain the layout and operation of the Hydro Power Plant with neat diagrams

A generating station which utilizes the potential energy of water at a high level for the generation of electrical energy is known as a hydro-electric power station.

Layout Diagram:



Main Components:

Reservoir: A large storage area, usually a dam, which holds water and controls the flow of water to the plant. It helps in storing water during periods of high inflow and releasing it during dry periods to ensure a steady supply.

Dam: A structure built to block a river, creating a reservoir or a head of water. The height of the dam creates potential energy by raising the water level.

Penstock: A large pipe or conduit that directs water from the intake to the turbines. The pressure of the water in the penstock increases as it moves downward.

Surge tank: A surge tank is a small reservoir or tank in which water level rises or falls to reduce the pressure swings in the penstock. A surge tank is located near the beginning of the penstock.

Turbine: A device that converts the kinetic and potential energy of falling or flowing water into mechanical energy. The water hits the turbine blades, causing the turbine to spin.

Draft Tube: After rotating the turbine, water is released into a downstream river or canal through the **draft tube**.

Generator: Connected to the turbine, the generator converts mechanical energy into electrical energy. As the turbine spins, it rotates the rotor inside the generator, inducing an electric current.

Transformer: This device steps up the voltage of the generated electricity for transmission over power lines. High-voltage electricity travels more efficiently over long distances.

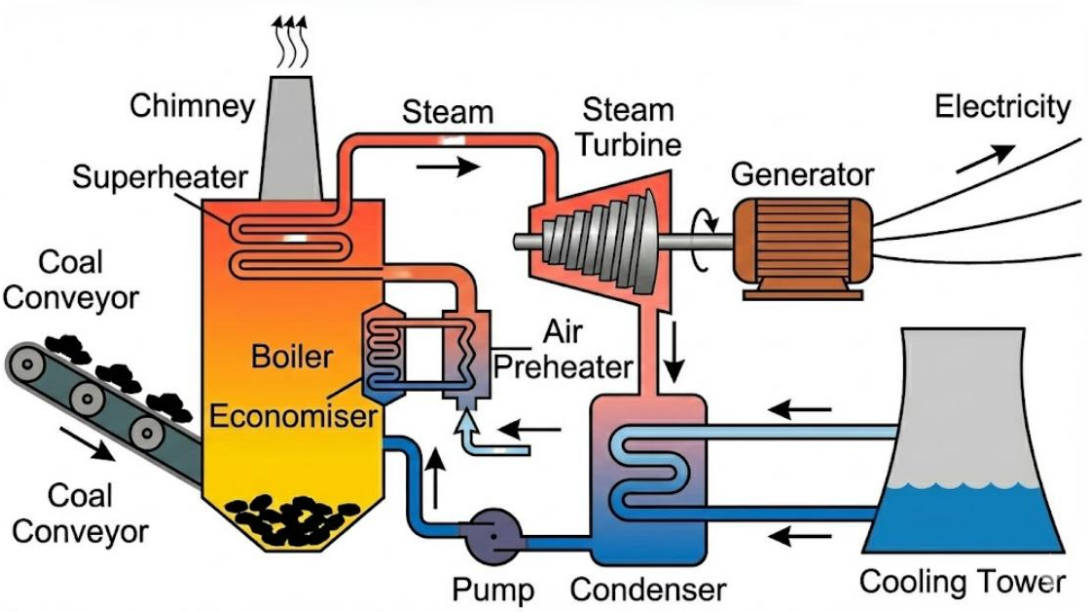
Working Operation:

A hydro power plant generates electricity by using the energy of flowing water. Water stored at a high level in a reservoir in a dam possesses potential energy. First, the water is released through a large pipe called a penstock, where its potential energy is converted into kinetic energy as it flows at high speed. The fast-moving water strikes the blades of a turbine, causing it to rotate and convert the kinetic energy of water into mechanical energy. The rotating turbine is coupled to an electric generator, and as the generator rotates, it produces electricity through electromagnetic induction principle. The generated electrical power is then stepped up using a transformer and transmitted to consumers through power transmission lines, while the used water is discharged back into the river through a draft tube.

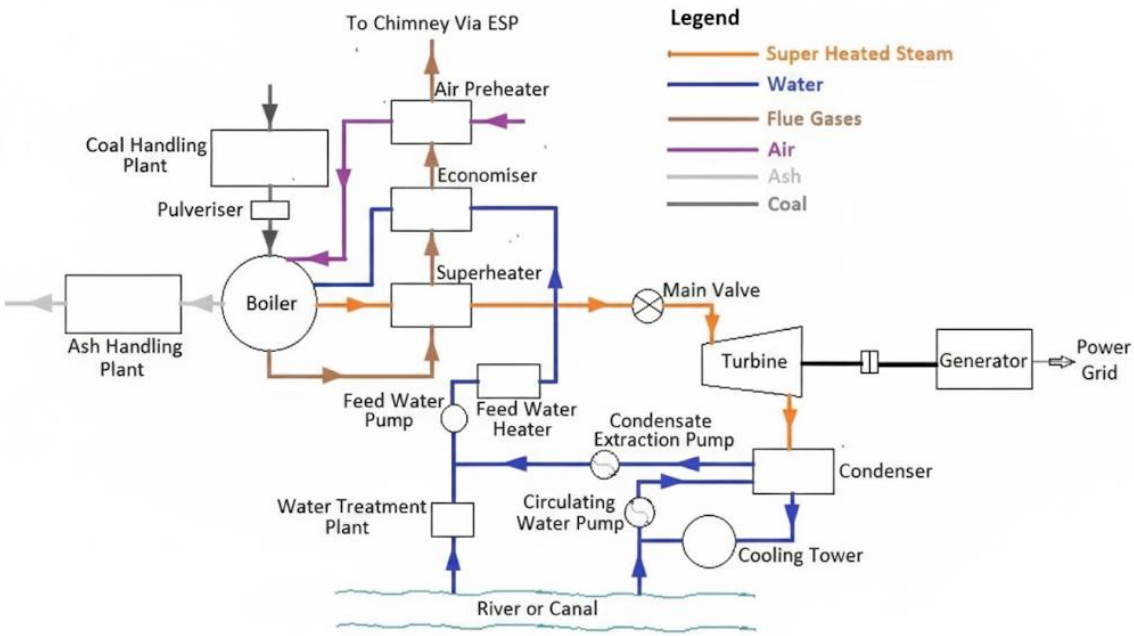
2. Explain the layout and operation of the Thermal Power Plant with neat diagrams

A thermal power plant generates electricity by using heat energy produced from burning fuel, usually coal, oil, or natural gas.

Layout Diagram:



(OR)



Main Components:

Coal: coal is the main fuel of the thermal power plant and is transported from coal mines to the generating station. Generally, bituminous coal or brown coal is used as fuel. The coal is crushed in small particles and then taken into pulverizer to make it in powdered form.

Boiler: A boiler is closed vessel in which water is converted into steam by utilising the heat of coal combustion. It converts steam at high temperature and pressure. This high-pressure steam is then sent to the turbine to rotate it and produce electricity.

Superheater. A superheater is a device which superheats the steam *i.e.*, it raises the temperature of steam above boiling point of water. The steam produced in the boiler is wet and is passed through a superheater where it is dried and superheated by the flue gases.

Economiser: An economiser is a device that uses the leftover heat from the boiler to warm the water before it enters the boiler. This helps the boiler use less fuel and work more efficiently.

Air preheater. An air preheater increases the temperature of the air supplied for coal burning by deriving heat from flue gases.

Steam turbine. The dry and superheated steam from the superheater is fed to the steam turbine through main valve. The heat energy of steam when passing over the blades of turbine is converted into mechanical energy.

Alternator. The steam turbine is coupled to an alternator. The alternator converts mechanical energy of turbine into electrical energy. The electrical output from the alternator is delivered to the bus bars through transformer.

Cooling arrangement. A condenser is a device that cools the steam coming from the turbine and changes it back into water. The water can then be reused in the boiler again. Water is drawn from a natural source of supply such as a river, canal or lake and is circulated through the condenser.

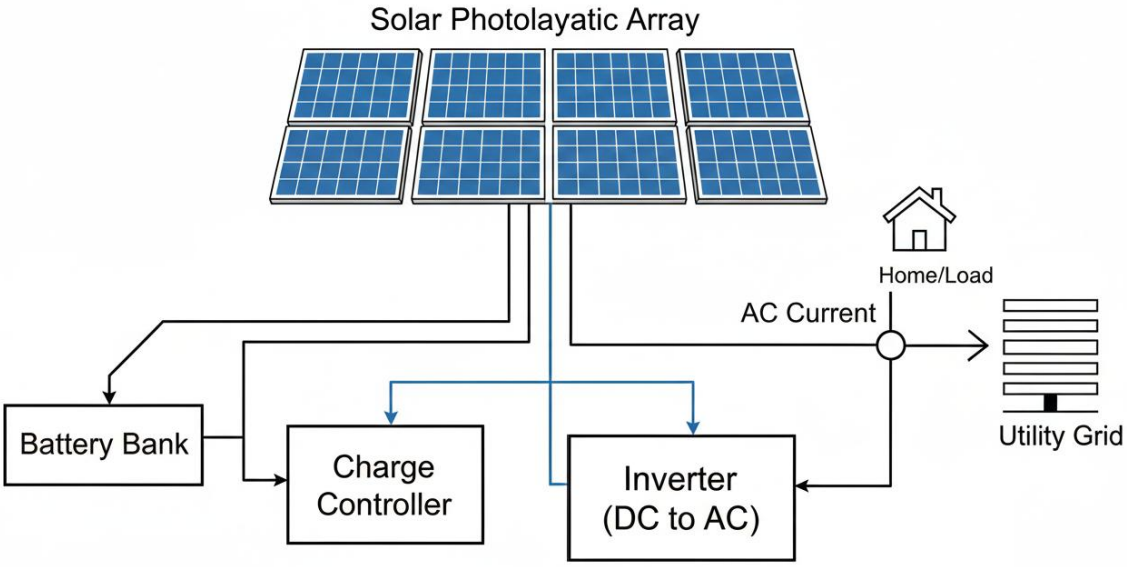
Working Operation:

A steam power plant works on the Rankine cycle. Coal is burned in the boiler to produce heat, and this heat is used to convert water into steam. Before the water enters the boiler, it is first warmed in the economiser using heat from flue gases. The steam produced in the boiler is then passed through the superheater to increase its temperature and pressure. This high temperature & high-pressure steam is sent to the steam turbine, where it rotates the turbine to produce mechanical energy. The turbine drives the alternator, which generates electrical energy. The steam is coming out from the turbine is cooled in the condenser and converted back into water to be reused.

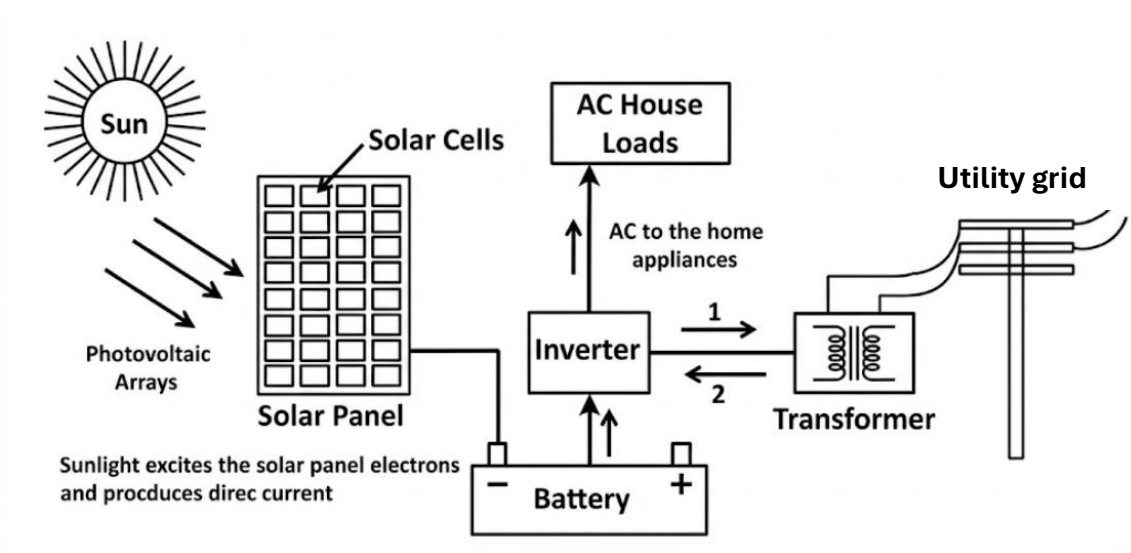
3. Explain the layout and operation of the Solar Power Plant with neat diagrams

Solar power generation is the process of **converting sunlight into electricity** using photovoltaic (PV) panels.

Layout of Solar PV Generation:



(OR)



Main Components:

Solar Photovoltaic Panels: A photovoltaic cell is a semiconductor device that generates electricity when sunlight falls on it. When sunlight hits on the surface of solar Photovoltaic cells, it releases electrons, and this movement of electrons produces electric current. This effect called Photovoltaic effect. The Power produced by the Solar Photovoltaic panels is DC.

Mounting structures: A mounting structure is a metal frame used to hold and support solar panels in place. It fixes the panels on a roof or on the ground and keeps them at the correct direction and angle to receive maximum sunlight.

Battery: The battery stores DC power produced by the solar panels during the day. This stored DC electricity can be used in the night time when the sunlight is not available, to provide a continuous power supply.

Charge controllers: A charge controller is a device used in a solar PV system to protect the battery from overcharging and Overdischarging.

Inverter: The electricity generated by solar panels is in DC form. Inverter is used to convert the DC power from the solar panels or battery into AC power.

Utility Grid: It is the main source of electricity provided by the Government electricity board. A utility grid is the network of power lines, transformers, and power stations that supply electricity to homes, schools, industries, and other places.

Net Meter: It is a bidirectional meter that measures electricity in two directions — the power taken from the grid and the power supplied to the grid. When the solar system generates more electricity than is needed, the excess is sent to the grid, and the meter records this surplus.

Working Operation:

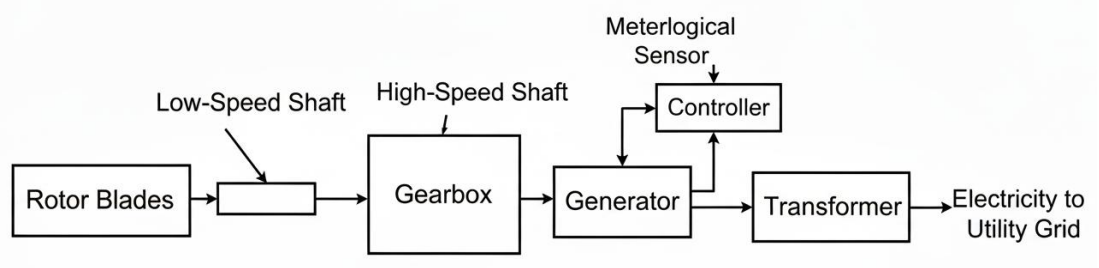
A solar power plant converts sunlight into electricity using solar panels (PV modules). When sunlight falls on the panels, they produce DC (direct current) power. This DC

power is sent to an inverter, which converts it into AC (alternating current) because most appliances and the power grid use AC. The generated AC power is then supplied to homes, industries, or the electrical grid. If the plant has a battery, extra solar energy during daytime is stored in the batteries and used later when sunlight is not available at night. The system also includes controllers and protection devices to monitor the power, prevent faults, and ensure safe operation.

4. Explain the layout and operation of the Wind Power Plant with neat diagrams

Wind power generation is the process of producing electricity using the energy of moving air (wind).

Layout of Wind power Generation:



Main Components

Rotor Blades:

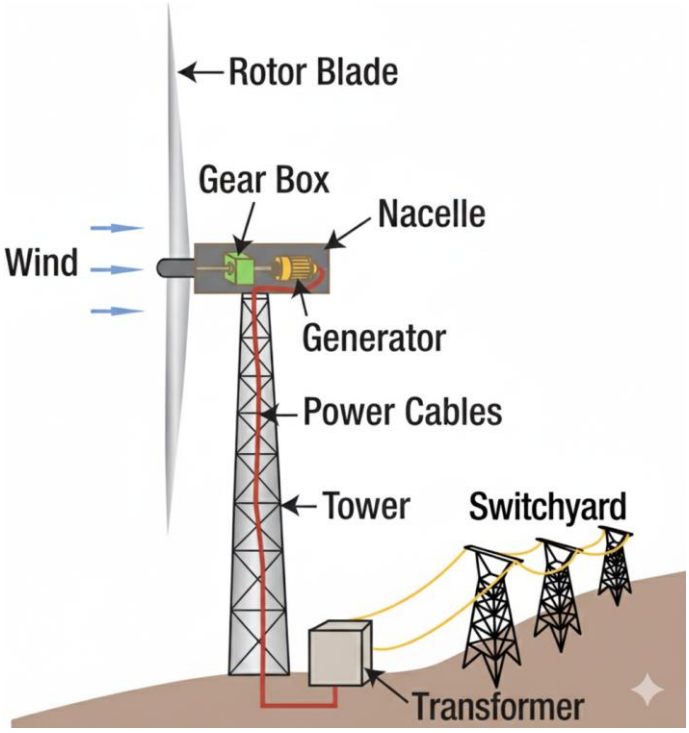
It captures kinetic energy from the wind and rotate like a propeller. The aerodynamic airfoil design helps maximize rotation even at low wind speeds.

Gear Box

It increases rotational speed from low-speed shaft to high-speed shaft—e.g., from 20–30 rpm up to ~1500 rpm, which is suitable for the generator.

Nacelle

It is the enclosed structure located at the top of the tower. The protective housing that contains mechanical components such as gearbox, shafts, and generator. Mounted on top of the tower.



Yaw System

It aligns the blades to face the wind direction. It works automatically using a wind direction sensor.

Tower

The tower is the tall support structure of a wind turbine. It holds the nacelle and blades high above the ground to capture stronger and more consistent winds.

Anemometer

Measures wind speed. Sends signals to the controller to start/stop turbine based on safe operating ranges.

Controller

The controller is an electronic system that starts the wind turbine when the wind speed is safe and stops it when the wind is too high or unsafe.

Generator

The generator converts the rotational mechanical energy from the turbine shaft into electrical energy.

Transformer: Steps up the voltage of the electricity generated by the wind turbine for efficient transmission to the grid.

Electrical Cables: Transmit electricity from the wind turbines to the transformer and then to the grid.

Working Operation:

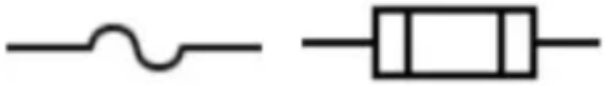
As the wind blows, a wind turbine converts the kinetic energy of the wind's motion into mechanical energy by the rotation of the rotor, and this mechanical energy is transmitted by the shaft to the generator through the gear train. The generator converts this mechanical energy into electrical energy, thereby generating electricity. A wind turbine is connected to the electricity network via a transformer located at the base of the tower. The transformer stepped up the voltage, and it send to the utility grid.

5. Explain the principle of operation of: i) Fuse ii) MCB

i) Fuse:

An electrical fuse is a safety device that operates to provide protection against the overflow of current in an electrical circuit. An electrical fuse is a metal wire or strip that melts when excess current flows through it.

Standard Fuse Symbols



The material used for fuse wires must have the following characteristics:

- Low melting point e.g., tin, lead.
- High conductivity e.g., silver, copper.
- Free from deterioration due to oxidation e.g., silver.

Working Principle of an Electric Fuse

The basic principle behind an electric fuse is the heating effect of electric current. As current passes through the thin fuse wire, heat is generated due to resistance. If the current surpasses the fuse's rated current limit, the generated heat increases the wire's temperature until it reaches its melting point. The fuse wire then melts and opens the circuit. This action ensures that excess current cannot flow through the protected circuit, preventing short circuits or overloading from harming appliances or causing fire hazards. After a fuse blows, it must be replaced with a new one.

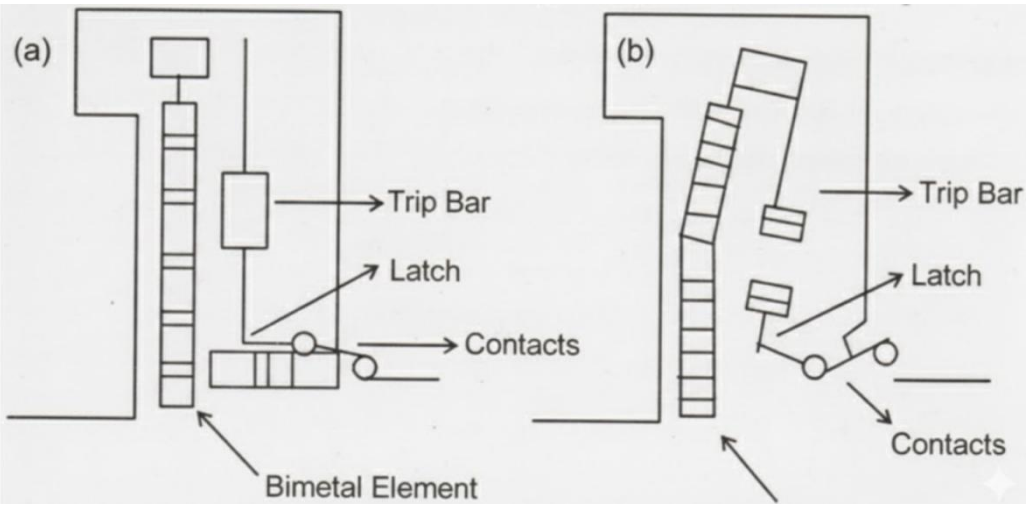
ii) Miniature circuit breaker (MCB)

A miniature circuit breaker (MCB) is an Electrical Switch that **automatically switches off** the electrical circuit during an abnormal condition of the network means an overload condition as well as a faulty condition.

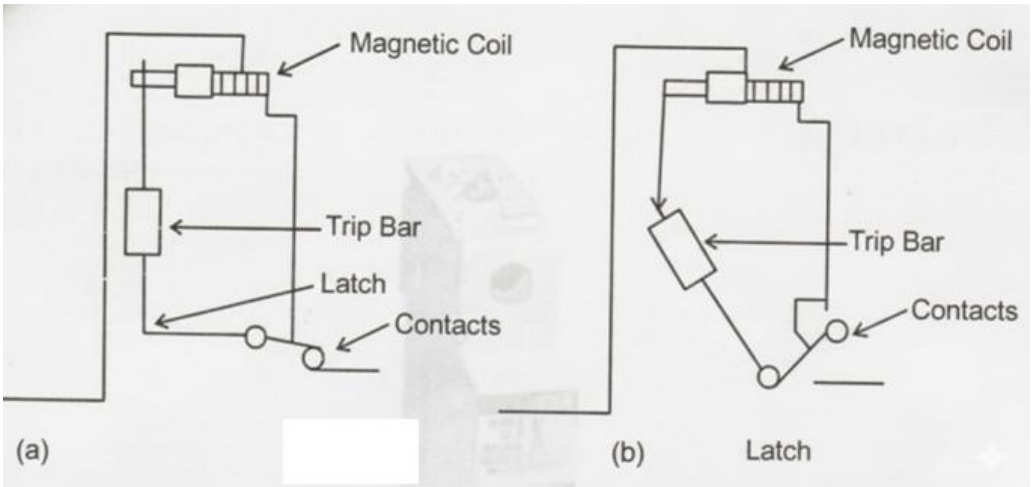
Working Principle of an MCB:

Normal Operating Condition: In standard operation, the MCB acts as a switch. Current flows from the incoming terminal, through the magnetic coil (solenoid) and the bimetallic strip, to the outgoing terminal, keeping the circuit closed.

1. **Overload Protection (Thermal Operation):** When a current slightly higher than the rated capacity flows through the MCB for an extended period, the bimetallic strip heats up and begins to bend. This bending eventually triggers a mechanical latch that releases the contacts, tripping the breaker and opening the circuit.



2. **Short-Circuit Protection (Magnetic Operation):** In the event of a short circuit, a massive surge of current flows through the solenoid. This creates a strong magnetic field that instantly pulls a plunger, which strikes the trip lever. This action opens the contacts almost instantaneously (within milliseconds) to prevent fire or damage.



Manual Reset: Once the fault (overload or short circuit) is cleared, the MCB can be manually switched back to the "ON" position to restore power.

6. What is electric shock and what are the precautions to be taken to prevent electric shock?

Electric shock

Electric shock occurs when an electric current passes through a person's body. It can cause pain, burns, muscle contractions, unconsciousness, or even death depending on the strength of the current. Electric shock usually occurs when someone touches a live electrical wire, damaged appliance, or wet electrical equipment.

Precautions to Prevent Electric Shock

✓ Do not touch electrical devices or switches with wet hands.

✓ Avoid using damaged wires, plugs, or appliances. Replace them immediately.

✓ Make sure electrical equipment and sockets are properly earthed (grounded).

✓ Use MCBs, fuses, and ELCBs (Earth Leakage Circuit Breakers) for protection.

✓ Wear rubber gloves and shoes when working with electricity.

✓ Do not overload power sockets with too many devices.

✓ Turn off the main power supply before repairing or touching electrical equipment.

✓ Always use insulated screw drivers, and line testers.

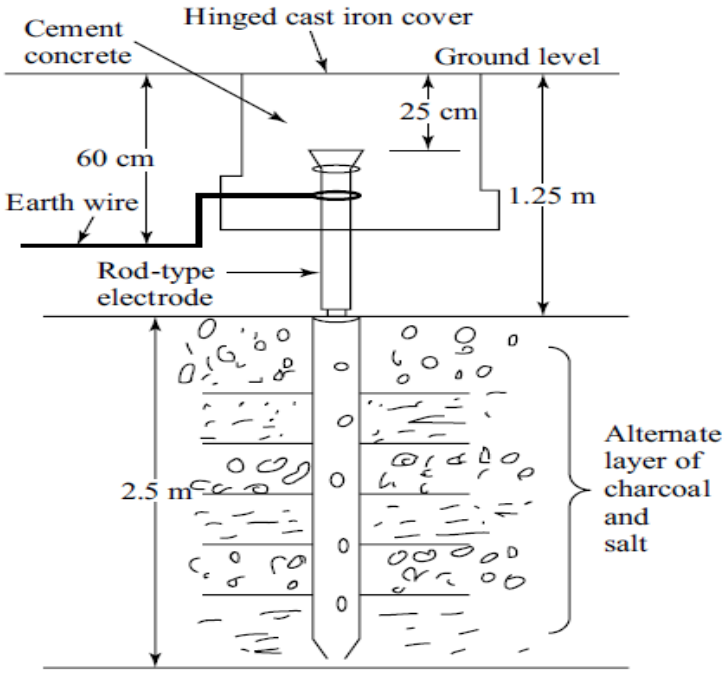
7. What is earthing? Explain different types of earthing.

Earthing

Earthing or grounding is the process of transferring and immediate discharge of electrical energy to the earth directly through a low resistance wire. When too much electricity flows due to a fault, the extra current is sent directly into the earth through this low-resistance path. Earthing is important in homes, industries, and buildings as it ensures that any fault current is safely discharged into the ground, reducing the risk of accidents.

Earthing Procedure

Earthing is done by connecting the electrically conductive part of an electrical equipment to the ground through earth plate or electrode. In general, galvanised iron is used as the material for earthing purpose. Before placing the earth electrode, the soil around is made highly conducting by adding some agents like common salt, i.e., NaCl, calcium chloride, sodium carbonate, soft coke, etc.



Types of Earthing

a. Plate Earthing

Plate earthing involves burying a metal plate, typically made of copper or galvanized iron, in the ground. The plate acts as an electrode and is connected to the electrical system.

b. Pipe Earthing

Pipe earthing utilizes a metal pipe, such as a galvanized iron pipe or a copper pipe, as the electrode. The pipe is driven vertically into the ground, and the electrical system is connected to it.

c. Rod Earthing

Rod earthing involves the use of a metal rod, usually made of copper or galvanized iron, driven vertically into the ground. The electrical system is connected to the rod.

d. Strip Earthing

Strip earthing employs a metal strip, such as a copper strip or a galvanized iron strip, buried horizontally in a trench. The strip acts as the electrode and is connected to the electrical system.

8. Problems on Electric Power Bill.

Example: A consumer uses a 10kW geezer, a 6kW electric furnace and five 100W bulbs for 15 hours. How many units (kWh) of electrical energy have been used and what would be the electricity bill?

Appliance	Count	Power Capacity (W)	Time (Hours)	Energy Consumption (Power × time) (kWh)
Geezer	1	1 × 10 kW	15	10 ×15 = 150 kWh
Electric furnace	1	1×6 kW	15	6 × 15 = 90 kWh
Bulb	5	5×100 = 500 W = 0.5 kW	15	0.5 × 15 = 7.5 kWh
Total Energy Consumed in (kWh)				**247.5 kWh (or) 247.5 Units**

1 unit costs Rs.2.5,
Then the total electricity bill= Total no.of. units × 1 unit cost

= 247.5 × 2.5 = **Rs.618.75/-**

Note:
Energy (Wh) = Power (W) × Time (hours)
1 kWh = 1 Unit

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* All the Best*